NAIMIRIS

A PROFIT-DRIVEN SMART FARMING MODEL FOR SCOTCH BONNET PEPPERS

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INDIVIDUAL THESIS

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ABSTRACT

AI-Driven Pest Monitoring and Control is dependent on information regarding the health status of crops, but the confusion arising from having the same leaf symptoms among different diseases makes disease detection hard. The current work analyses the AI-Driven Pest Monitoring and Control System, one of the individual components of NAIMIRIS, which is a smart farming application aimed at generating profits using Scotch Bonnet pepper. The system employs the use of EfficientNet-B0 Convolutional Neural Network coded in PyTorch for the classification of the input image into any of the six classes or healthy class. The six disease classes include bacterial spot, Cercospora leaf spot, leaf curl, powdery mildew, anthracnose, broad bean wilt virus 2 infection and healthy. The updated preparation pipeline includes 4,473 images kept and applies grouped training, validation, and test sets to minimize duplicate data leakage. The mobile prediction pipeline is stated as having been developed at Progress Presentation 2, while the recommendation component is considered to be a different, source-based knowledge base. Severity levels of the diseases and automatic pesticide application rates fall out of scope of the implemented classifier. Final performance on the holdout dataset, local validation in the field and the results of stakeholders' implementation are not confirmed yet; hence, no accuracy or agronomic benefits could be reported quantitatively. The contribution is a reproducible screening pipeline with detailed class mapping, deployment conditions and guidance. However, its applicability to the case of Sri Lanka's Nai Miris needs confirmation since the training data includes public chilli and pepper images (including controlled environment images of Capsicum annuum) instead of an independent verification of the local Scotch Bonnet dataset.

Keywords:
[Nai Miris, EfficientNet-B0, disease classification, transfer learning, decision support]

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
API	Application Programming Interface
BBWV2	Broad Bean Wilt Virus 2
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
DOA	Department of Agriculture
DSR	Design Science Research
EXIF	Exchangeable Image File Format
F1	Harmonic mean of precision and recall
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
IPM	Integrated Pest Management
JSON	JavaScript Object Notation
PP2	Progress Presentation 2
RGB	Red Green Blue
SHA	Secure Hash Algorithm
UI	User Interface

1 INTRODUCTION

1.1 Overall project context

NAIMIRIS is an intelligent farming project aimed at aiding profit-focused cultivation of the Scotch Bonnet pepper crop. The project involves precultivation analysis, farming assistance with technological support for crops and their health monitoring in one mobile application. In the PP2 presentation, the target audience of the platform consists of small-scale farmers and potential agricultural investors who will be able to use the platform to understand the information about the land, resources, growth and health of the crops, not just to have fragmented observations [1].

The specific component analyzed in this thesis is the AI-Driven Pest Monitoring and Control System. It adds crop health monitoring by image recognition to the platform as a whole. A farmer takes an image of the leaf, the backend classifies the image using a trained classifier, and the application gives the predicted category along with the confidence score. The management information is processed in the recommendation layer.

1.2 Individual component context

This particular component solves the problem of receiving a timely evaluation of visible symptoms on leaves. The implemented AI task is one-shot classification into multiple classes using the EfficientNet-B0 model from PyTorch. Classification classifies images into one of seven classes: healthy, bacterial spot, Cercospora leaf spot, leaf curl, powdery mildew, anthracnose, or BBWV2 infection. Class keys and their order belong to the model contract and cannot be modified via translation or renaming of labels in the interface [2].

The chosen name of this particular component includes the words pest monitoring and control in the general meaning of protection of plants. At the same time, the current version of the model identifies only classes related to the disease and healthy condition of leaves but does not identify insects or count pests or even use the sprayer. Control here implies recommendations based on screening results.

1.3 Problem statement

There are many reasons why pepper leaves could develop spots, curling, yellowing, and other visual abnormalities. Such an image could contain symptoms of disease, insect damage, environmental stress, or any other factor. Therefore, in order to be helpful to the farmer, such a system should not imply that the image proves the lab-confirmed presence of the particular condition.

The actual research problem here is not just classification but also clear communication of what action should come after the recognition. Individual research question is to create a replicable image classification workflow which would screen the supported conditions in the NAIMIRIS application. This should involve not only proper pre-processing and class mapping but also clear separation of prediction confidence from the actual performance of the model.

1.4 Research gap

The study of deep learning has shown that it is possible to identify disease classes of plants based on leaf images [3], [4]. EfficientNet is an architecture known to facilitate efficient visual feature learning [5]. The problem solved in this project is not about the development of a novel convolutional neural network. This project is focused on the creation of the screening module of crop health as a part of NAIMIRIS process.

The initial idea was oriented at the identification of the severity of the disease and dosages of herbicides. The updated classifier does not quantify the area of affected leaves nor provides the labels for severity. The contribution of this work is thus limited to the issues of classification, reproducibility, and recommendation design.

1.5 Aim and objectives

The objective is to design and assess an AI-based leaf image screening system for NAIMIRIS that categorizes supported disease-related conditions and conveys results using a mobile workflow with traceable management guidance.

The specific objectives are:

- Gather and annotate a public domain image set comprising six disease categories and one healthy category, preserving image sources and licences.

- Maintain standardised image processing, duplicate elimination and fixed image partitions in groups for repeatable experimentation.
- Modify an EfficientNet-B0 ImageNet classifier to classify images into seven categories and incorporate support for interrupted training via checkpoints.
- Integrate image upload, backend classification and category/output confidence display into the NAIMIRIS app.
- Detail guidance in category management, uncertainty management and clear boundaries regarding disease classification and chemical advice.
- Assess image classification effectiveness, software operation and usability with distinct evidence.

The goals are based on the changed technical scope. Severity assessment and automatic calculation of doses are left for future extension. The exclusion of severity assessment and automatic dose calculation from the current scope does not mean the original goals have been accomplished; this is reflected in Chapter 9.

1.6 Research questions or hypotheses

RQ1 examines the classification performance of the seven-class EfficientNet-B0 classifier on leaf images from an external source. RQ2 evaluates the potential of the grouped data preparation and checkpointing method to provide a replicable experiment that does not contain any duplications and contradictions. RQ3 evaluates the ability of the application to present the results while preserving all information regarding the preprocessing procedure and class mapping used in the model. RQ4 examines the potential for category guidance and uncertainty messages to guide crop health decisions.

1.7 Scope limitations and assumptions

The input is a photograph expected to contain either one leaf or a distinctively affected area on the leaf. The output from the classifier is one of the seven categories that have been trained. In some cases, the application may give alternative classifications and require another photograph if the classification is ambiguous. It cannot prove that the whole plant or field is healthy on the basis of one classification in the healthy category.

Currently, the application's scope excludes such things as insect localization, mites, thrips, nutritional deficiencies, multi-label diseases, unknown disease identification, pathogen strain confirmation, disease severity, and automated pesticide dosing. An image that is outside of the closed set classification may receive a high score. Therefore, the confidence thresholds are merely communication controls.

1.8 Individual contribution

The individual element encompasses the classification, organization of disease categories, the processing and transferring of images for training, the need for classifier integration, and category guidance design. This is different from the other aspects of the project, such as land analysis, decision making about finances, and growth stage tracking. The shared app infrastructure is recognized as team effort and not individual work.

1.9 Thesis structure

Chapter 2 gives the review of literature and agricultural guides. Chapter 3 details on research methodology, data pre-processing, and configuration and evaluation of model. Chapter 4 details on requirements, architectural design, and disease-guidance design. Chapter 5 details on the implementation process, integration agreements, and test cases. Chapter 6 gives all the results obtained and pending validations. Chapter 7 gives the contributions and limitations. Chapter 8 covers the aspects of commercialization and sustainability, and chapter 9 gives the conclusion with the objective achievement and future direction.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

Web-based sources were accessed on 10 October 2026. This is an illustrative review on plant image classification, transfer learning using EfficientNet, publicly available chilli data, and uncertainty communication and pepper disease management. This review uses primary literature, documentation for the framework, data set literature, and agricultural guidance by government or university organizations. This review does not make claims of the extensive search and screening process of a systematic review.

2.2 Domain and theoretical foundations

2.2.1 Computer vision for plant health

Computer vision encodes an image numerically and learns patterns in correlation with labelled classes. For application in plant disease diagnosis, leaf color, texture, and shape may offer valuable information, but the image is merely a depiction of visual observations. Similar symptoms for various diseases weaken an inference based solely on image. The decision support system therefore suggests a potential disease rather than identifying a specific pathogen.

Classification, detection, and segmentation are different problems. Classification is assigning a class to an image, while detection deals with the location of objects or regions, and segmentation involves labelling parts of the image. The current design of NAIMIRIS employs classification. One cannot infer any bounding box, damaged area percentage, or pest count from seven class scores.

2.2.2 Deep learning and CNN based classification

EfficientNet is the name for a set of convolutional neural networks that combine well network depth, width, and image size [5]. Architecture B0 is selected as the feature extractor for this task. Transfer learning on ImageNet will provide the initial representation of images, while a 7-layer output layer will adjust the architecture according to the selected labels.

2.2.3 Classification alternatives and localization

Object detection could prove useful in cases where the job involves locating multiple insects or areas of infection in an image. In terms of lesion area estimation, the use of segmentation could prove effective depending on the availability of proper masks. The present source and target formats are based on categorization of images; hence the appropriateness of classification.

2.2.4 Confidence uncertainty and severity

Softmax transforms the seven output logits into probabilities that add up to one. The probability with the highest value determines the winner, although this is not equal to accuracy that would have been computed using a labeled test data set. Neural network predictions tend to be miscalibrated, according to Guo et al. [6]. Confidence is not a measure of the damaged area.

2.2.5 Decision support for treatment

The classification system and the management knowledge base have distinct roles to play. While the model can identify visual classes, the knowledge base offers descriptions, preventive actions and recommendations. Such a distinction ensures that the recommendation guidelines can be updated or reviewed without retraining the neural network. The recommendation also remains traceable in terms of origin and version.

Sri Lankan agricultural recommendations and university pepper management literature are used as a source of general sanitation, monitoring and management recommendations [7], [8], [9], [10]. The presence of these sources does not confirm the legitimacy of each chemical substance in all possible applications. The tool does not consider the confidence factor when calculating dosages.

2.3 Related research

Accuracy of 99.35% was achieved for leaf images by Mohanty, Hughes, and Salathe using deep learning with 54,306 leaves [3]. This study proves the viability of using images for classification, but the achievement is limited to that of their particular crops, categories and experimental setting. It cannot be extrapolated for comparison to the seven-category Nai Miris application as an accuracy measure.

A convolutional neural network design for plant disease diagnosis was proposed by Ferentinos and 99.53% of success rate was reported [4]. The value in the paper is the use of learned visual representation among multiple plant-condition categories. However, different data and experimental setting do not allow for direct ranking relative to NAIMIRIS without common evaluation set.

EfficientNet model architecture is the basis of the selected model [5]. The efficiency considerations are valid for cases when there is a scarcity of training data and inference computational resources. Calibrations studies add one more point that must be considered, as the model ranking its classes correctly does not guarantee the numerical probability it provides [6].

2.4 Existing systems products and techniques

Table 2.1 compares the relevant evidence and its role in the component. The comparison focuses on transferable design lessons rather than presenting incomparable numerical scores as a single league table.

Table 2.1 Related work and relevance to the individual component

Work or source	Main contribution	Relevance and limitation
Mohanty et al. [3]	Deep learning for labelled plant-health images	Supports classification feasibility; dataset-specific performance is not local field accuracy.
Ferentinos [4]	CNN-based plant and disease recognition	Supports transfer of image-recognition methods; different classes prevent direct performance comparison.
Tan and Le [5]	EfficientNet architecture family	Provides the selected backbone; no Nai Miris-specific validation is implied.
Guo et al. [6]	Neural-network calibration analysis	Supports careful interpretation of softmax scores and validation of thresholds.
CHILLI and mobile anthracnose datasets [11], [12]	Public labelled chilli photographs	Provide selected source categories; labels and crop identity need critical review.
COLD dataset [13]	Raw and augmented chilli categories	Supplies selected raw images; non-disease categories are excluded from the six-disease design.

Work or source	Main contribution	Relevance and limitation
BBWV2 dataset [14]	Laboratory pepper infection and stress imagery	Supports the BBWV2 source label; creates a strong environment and crop-transfer limitation.
Agricultural guidance [7], [8], [9], [10]	Symptoms and disease management	Supports the knowledge base; advice must be reviewed for the local application.

2.5 Gap synthesis and conceptual framework

Figure 2.1 shows the conceptual framework. The review supports a framework with three linked responsibilities: reproducible image classification, faithful application integration and traceable management information. Public research supports the feasibility of the first responsibility, while official guidance supports the content of the third. The component must connect them without hiding uncertainty or claiming diagnostic confirmation.

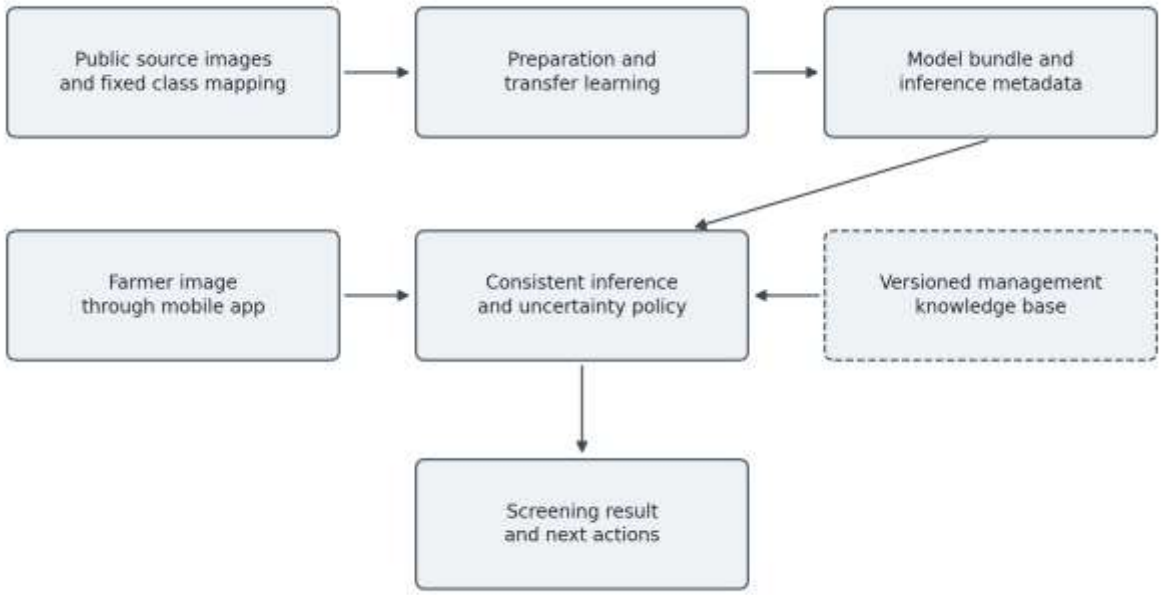


Figure 2.1 Conceptual framework for classification and decision support

3 RESEARCH METHODOLOGY

3.1 Research design

This research employs a Design Science Research approach: an artefact is designed to solve a real-life crop health information problem, and the artefact is evaluated using technical and end-user evidence. Hevner et al. explain the connection between developing an artefact and the evaluation of its usefulness [15]. In this thesis, the artefact consists of the classifier, deployment and results workflow for farmers.

Evidence is divided into documented configuration, documented development verification, documented application integration and empirical performance measure. The first three provide the foundation for what has been designed or documented as functioning. They are not a substitute for an evaluation of the artefact using a test set or by the farmers.

3.2 Data participants or study materials

The modified workflow utilizes four open sources as stated in Table 3.1. This workflow involves reading images from the big CHILLI database, the chosen mobile captured anthracnose images, chosen raw COLD images, and the healthy/BBWV2 part of the lab database. Overwatering, nutrients deficient, and both mites & thrips are removed. The substitute notebook keeps track of identifiers and filterings [2].

Table 3.1 Public data sources and selected categories

Source	Selection for this workflow	Provenance considerations
CHILLI on Kaggle [11]	Original bacterial spot, Cercospora, curl-virus and healthy images	Pre-augmented folders are excluded; source reports CC BY-NC-SA 4.0.
Mobile-captured anthracnose on Kaggle [12]	Anthracnose and healthy images	Source annotations require review; workflow records CC0.
COLD through Project-AgML [13]	Raw Cercospora, powdery mildew and healthy images	Other classes are excluded; dataset card lists CC BY 4.0.
BBWV2 and overwatering dataset [14]	BBWV2 and healthy images only	Controlled Capsicum annuum imagery; flooding stress is excluded.

Table 3.2 shows the number of images prior to duplicate and conflict elimination. The sum of these numbers comes to 4,507 images. The common preparation record has a number of 4,473 images that have been kept after their process, which results in a drop in 34 images. This number is a preparation number reported in the development record [2].

Table 3.2 Selected images before duplicate and conflict processing

Model category	Selected images	Interpretation
healthy	2,197	Additional healthy category, not a disease
bacterial_spot	156	Source-labelled bacterial spot
cercospora_leaf_spot	332	Source-labelled Cercospora leaf spot
leaf_curl	423	Source label includes Curl Virus
powdery_mildew	102	Selected raw mildew images
anthracnose	162	Source-labelled anthracnose
bbwv2	1,135	Source-labelled BBWV2 infection
Total	4,507	Before duplicate and conflict processing

3.3 Experimental or study procedure

This workflow starts off by checking if there is an existing validated prepared cache in Google Drive. In case it is, it reuses the prepared images and manifest without having to download the source archives again. In case it does not exist, it then downloads the desired source images, validates their format, and prepares the desired categories. Refusal downloads end up falling back to the manual sources and not other datasets [2].

Orientation is done through EXIF data, conversion into RGB, cropping the images such that they fit into a 224 by 224 pixel square while maintaining the aspect ratio, and filling the remaining area with the value RGB of 128. Pixel hashes and perceptual hashes help in handling duplicate images. Images similar and images with the same original-photo identification are put into the same group.

The groups are stratified by class with random seed 42. About 70% of groups go into the training set while the other 30% are equally split into validation and test sets. Since the group sizes are different, these proportions are not exact in terms of the percentage of images. The manifest is the ultimate authority on the per-class partition sizes. The code verifies that a group does not appear in more than one partition.

Training augmentation is limited to the training partition only and consists of horizontal flip, rotation up to 15 degrees, and color augmentation. For evaluation, the prepared image without any random augmentation is used. Tensor conversion and ImageNet channel normalization are used in both paths. Mean: [0.485, 0.456, 0.406]; standard deviation: [0.229, 0.224, 0.225].

EfficientNet-B0 with ImageNet pretraining and with a seven-node linear layer in its output is employed in this experiment. The model configuration has three epochs of training heads before fine tuning for a total of 30 epochs. Thirty epochs are the configured training, not an assurance that all 30 have been completed. The main parameters of this configuration are provided in Table 3.3.

Table 3.3 Training configuration documented in the revised notebook

Setting	Configuration
Architecture and framework	torchvision EfficientNet-B0 in PyTorch
Initial weights	ImageNet pretrained weights
Model input and output	RGB 224 by 224; seven logits
Seed and batch size	42 and 32
Head training	First 3 configured epochs; feature parameters frozen
Total schedule	30 configured epochs
Optimizer	AdamW; weight decay 0.0001

Setting	Configuration
Learning rates	Head 0.001; fine-tuning 0.0001
Training loss	Cross entropy weighted by normalised inverse square root of training-class counts
Validation loss	Unweighted cross entropy
Scheduler	ReduceLROnPlateau on validation macro-F1; factor 0.5; patience 2
Selection criterion	Best validation macro-F1
Stability and resource controls	Gradient clipping at norm 1.0; AMP on CUDA
Session management	4.5-hour budget with 10-minute saving reserve

The held-out test set can only be tested once the training process has been completed. The checkpoint with the best validation performance is then recovered to generate predictions, a classification report and the model metadata. Results from the test set should never be reused to make setting selections. Figure 3.1 provides an overview of this separation.

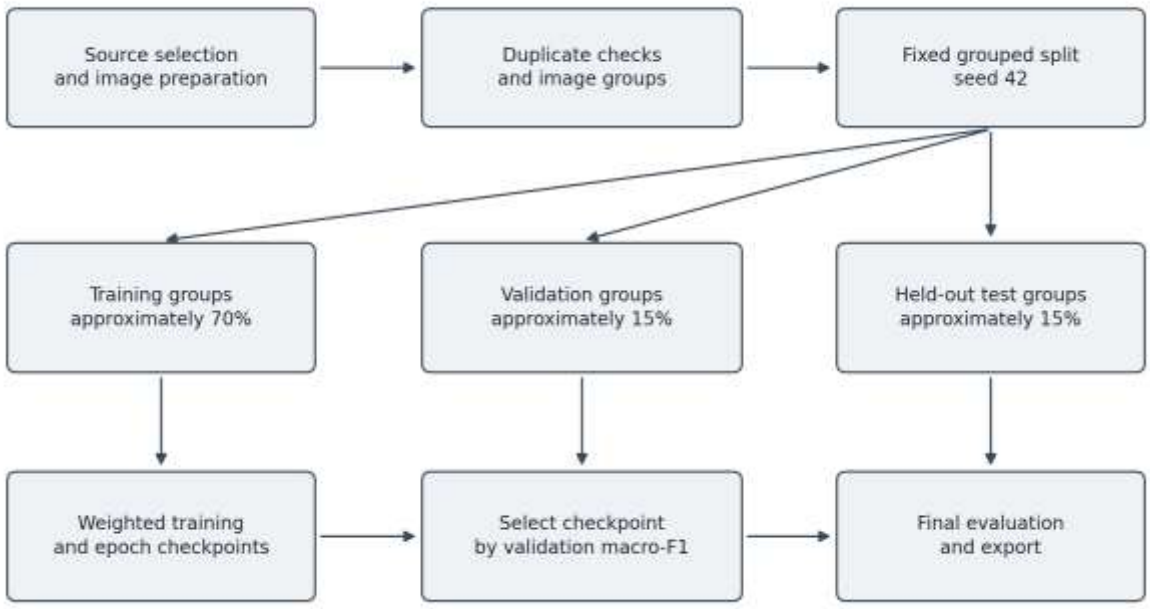


Figure 3.1 Grouped preparation training and evaluation workflow

3.4 Measures and evaluation metrics

Classifier performance is assessed using the class labels from the ground truth. The metric for accuracy refers to the number of correct classifications. Precision refers to the correctness of predicted instances for a certain class, recall refers to the ability to retrieve the positive cases of a certain class, and F1 score is the measure that combines both recall and precision.

Table 3.4 Evaluation measures and their interpretation

Measure	Definition or evidence	Purpose
Accuracy	Correct predictions divided by test images	Overall classification correctness
Per-class precision	TP divided by TP plus FP	Reliability of a category prediction
Per-class recall	TP divided by TP plus FN	Ability to recover examples of a category
Per-class F1	2PR divided by P plus R	Balance between precision and recall
Macro-F1	Mean of the seven class F1 values	Equal attention to each category
Confusion matrix	True category against predicted category	Identification of repeated confusions
Response time	Timed inference and complete request duration	Separation of model and application delay
Uncertainty behaviour	Accepted and deferred outcomes on labelled and unsupported images	Assessment of decision thresholds

Object-detection mAP is not considered the primary evaluation metric since this model doesn’t provide bounding boxes for objects. The confidence score is provided by each prediction and is not called test accuracy at all. The provisional top-score value is 0.70 and the top-two-margin one is 0.15. They are only candidates for validation and not operational assurances.

3.5 Data analysis

Analysis starts from the manifest and the class distribution of the manifest. It is necessary to differentiate between final image counts and group counts, and also the results from individual sources are analyzed since the source itself could have

correlation with the labels of the diseases. Accuracy from the total results could hide bad classification performance for minority classes or dependence on specific backgrounds.

3.6 Ethics privacy security and responsible AI

Data citation and licensing are essential to proper data use. Original sources can still be identified after harmonization of categories. The presence of non-commercial and/or share-alike licenses necessitates prior review for a paid deployment; availability for downloading is no guarantee of unlimited commercial use. Source attributions are not to be treated as laboratory-proven local diagnoses.

Pictures uploaded must be managed in accordance with a specific retention schedule. It is not appropriate to keep personal metadata and geographical data without a specific need. Authentication, transport security, and access controls belong to the application design and claims need verification. Consent for farmer-based studies will be necessary in the future.

3.7 Reproducibility and research artefacts

The experiment folder is NAIMIRIS_SIX_DISEASES_V2. The prepared cache, manifest, audit files and checkpoint metadata ensure the mapping between the images, labels and model. The new set of classes is different from the previous set containing only mites and thrips, therefore the old checkpoints are not interchangeable. The source version and COLD revision number are logged by the notebook [2].

The artefacts that should be present in the expected evaluation include classification_report.csv, test_predictions.csv, metrics.json, inference_config.json and inference.pt. The deployment name pest_detector.pt can be used to refer to the exported model bundle, however changing the name does not alter the architecture or keys inside the file.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements distinguish the main process of classification from the process of recommendations and validation. Table 4.1 shows the scope of the functionality. A requirement is an expression of the desired behaviour, and not an accomplished test. The evidence status will be evaluated separately in chapter 6.

Table 4.1 Functional requirements

ID	Requirement
FR1	Allow capture or gallery upload of a leaf image in the mobile application.
FR2	Validate the file and submit an analysis request to the backend.
FR3	Apply the model's documented RGB conversion, padding and normalisation.
FR4	Produce seven-category classification scores using EfficientNet-B0.
FR5	Preserve the fixed class-index mapping and display the original image, category and score.
FR6	Apply a configurable uncertainty policy and allow another image to be submitted.
FR7	Retrieve source-based management guidance from a separate category knowledge base.
FR8	Present healthy and viral results without inappropriate chemical-cure claims.
FR9	Return understandable errors when validation, model loading or communication fails.
FR10	Record model and guidance versions where analysis history is implemented.

Table 4.2 Nonfunctional requirements

Area	Design requirement	Verification approach
Correctness	Consistent class mapping and preprocessing	Known-index fixtures and tensor-parity checks
Performance	Practical response time on the deployment device	Record inference and end-to-end timings
Usability	Clear image input, progress and result states	Representative user tasks

Area	Design requirement	Verification approach
Reliability	Controlled failure when the model or service is unavailable	Failure-path tests
Security and privacy	Protected access and deliberate image retention	Configuration and access-control checks
Maintainability	Separate model, mapping and guidance versions	Deployment and rollback review
Scalability	Resource use appropriate to concurrent requests	Load tests before wider operation

4.2 Component architecture within the complete solution

The component works as part of the NAIMIRIS mobile application along with the pre-analysis and farming advisory services. The component interface takes an image and provides a structured response about the crop health status. The mobile application is in charge of capturing, previewing, uploading, and displaying the image, while the backend is responsible for validating, making predictions based on the model, and constructing the result.

Figure 4.1 depicts the architecture of the component within the larger NAIMIRIS application.

Figure 4.1 illustrates the architecture of the component within the broader NAIMIRIS application. The classification service will load the model once when starting up and will run the model in evaluation mode for every request. The metadata contract provides the order of the classes and the image transformation. Guidance service will retrieve the text by the stable class key. Persistent history, if used, will have to link the result with the user and request and the model it was generated with.

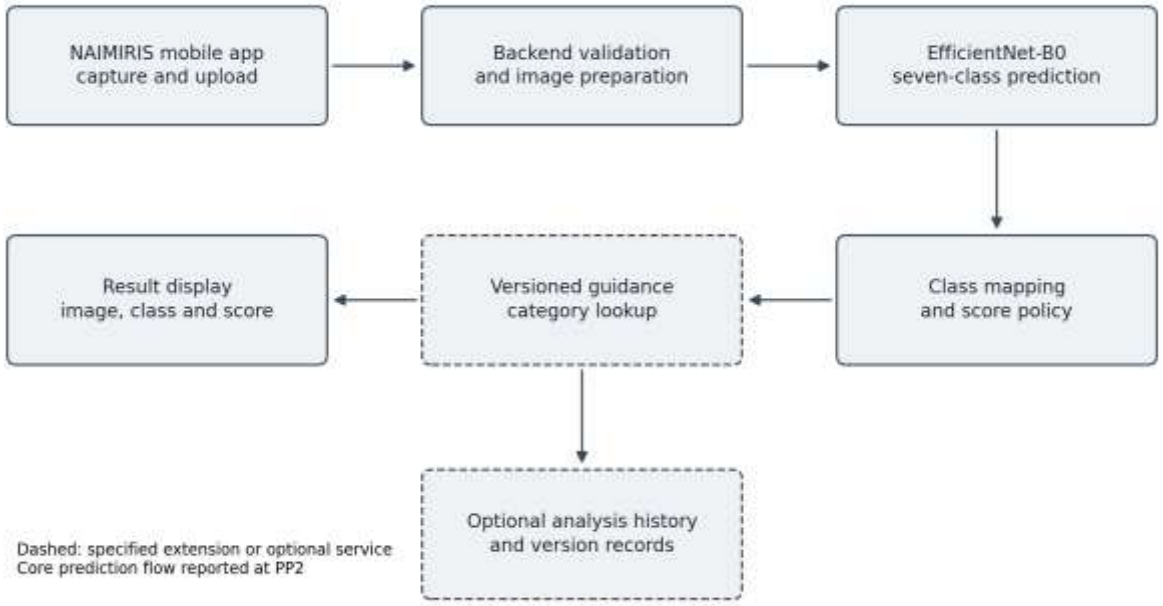


Figure 4.1 Revised crop health component architecture

4.3 Design alternatives and decisions

Table 4.3 records the main decisions. These are justified by the task and available annotations, not by an unperformed performance comparison. A future baseline experiment can test whether a different classifier improves the accuracy-resource tradeoff.

Table 4.3 Design alternatives and selected approach

Decision	Selected approach	Reason and limit
Visual task	Single-image classification	Matches category labels; does not localise symptoms.
Architecture	EfficientNet-B0 transfer learning	Established feature extractor; application performance still requires measurement.
Image geometry	Aspect-preserving fit and grey padding	Matches revised training preparation and export metadata.
Advice generation	Versioned category knowledge base	Separates agricultural content from neural-network scores.
Uncertain results	Score and margin policy	Supports cautious presentation; not an unknown-image detector.

Decision	Selected approach	Reason and limit
Execution	Backend inference	Matches PP2 prediction flow; host capacity is not yet benchmarked.
Severity and dosage	Deferred	No corresponding trained output or validated calculation exists in the classifier.

4.4 Detailed design

4.4.1 Image capture and upload

The interface offers options for the camera and the gallery path, an image preview, and an analysis step. The user is required to take a picture of one damaged leaf in good lighting conditions and ensure the appropriate part remains visible. The interface tells the user that a picture of fruit or overlapping leaves does not belong to the classifier's training scope.

4.4.2 Image preprocessing

The updated preprocessing contract of the notebook has preference over general pretrained-model examples. It includes EXIF orientation fix, RGB conversion, and 224 by 224 pixel aspect-preserving contain with center grey padding. The tensor is then normalized using the channel statistics of ImageNet from Chapter 3. Training-only augmentations are not applied during inference.

However, the common developer prompt suggested another preprocessing path, which included Resize(256) and then CenterCrop(224). This is contradictory to the training and export code updated in [2]. The unified deployment approach applies the preprocessing transformation of the notebook unless the real metadata of the checkpoint tells otherwise. The consistency check of the notebook and backend tensors of the same image is performed first before testing model performance.

4.4.3 Disease classification and class mapping

The seven class indices are set as per Table 4.4. The user-readable labels modify the language for the NAIMIRIS environment while keeping the machine-learned class indices the same. The system is expected to append 'possible' or 'suspected' to disease

results when necessary. The display label does not guarantee that all the images used are those of Scotch Bonnet.

Table 4.4 Exact model mapping and Nai Miris display names

Index	Stable model key	User-facing name
0	healthy	Healthy-looking Nai Miris leaf
1	bacterial_spot	Bacterial spot of Nai Miris
2	cercospora_leaf_spot	Cercospora leaf spot of Nai Miris
3	leaf_curl	Suspected leaf curl disease of Nai Miris
4	powdery_mildew	Powdery mildew of Nai Miris
5	anthracnose	Suspected anthracnose of Nai Miris
6	bbwv2	Suspected broad bean wilt virus 2 infection in Nai Miris

4.4.4 Uncertainty handling and scope control

This policy is to postpone the decision in cases where the highest score is less than 0.70 or where the difference between the top score and the second score is less than 0.15. This is followed by a request for a better image, which could involve showing some of the top options.

4.4.5 Treatment recommendation

The knowledge base assigns each key from the stable category to its corresponding description, symptom description, action to be taken immediately, information regarding how it can be prevented, sources of the advice and the date of review. The advice is extracted using the rules and not the classifier. The healthy and uncertain categories have no disease-specific chemical advice.

Table 4.5 Category-specific management focus

Category	Management emphasis	Important boundary
Healthy	Continue inspection and balanced crop management	No whole-plant health guarantee or routine pesticide instruction.

Category	Management emphasis	Important boundary
Bacterial spot	Clean planting material, dry handling, tool hygiene and reduced leaf splash	Chemical efficacy is not assured by the model [8].
Cercospora leaf spot	Remove infected debris, improve airflow and obtain local leaf-spot advice	Similar lesions need confirmation [10].
Leaf curl	Review possible causes, inspect vectors and use healthy planting material	Image appearance does not prove a particular virus.
Powdery mildew	Inspect leaf undersides and obtain early management advice	Sanitation alone may be insufficient [9].
Anthracnose	Remove infection sources, improve drainage and inspect fruit separately	A leaf classifier is not a validated fruit detector [7], [10].
BBWV2	Expert confirmation, sanitation and review of aphid-related spread	Retain the virus label; no pesticide cure is implied [14], [17].

4.4.6 Storage security and traceability

Analysis record may contain Request Identifier, Stable Class Key, Scores, Diagnosis State, Model Version, and Guidance Version. Storage of images is not mandatory and should be done only based on the purpose of the application and consent arrangements. Logging should facilitate troubleshooting without exposing users' images or credentials.

4.5 Standards and professional practice

Professional practice is evident in terms of traceable sources, stability of interfaces, versioning, reproducible data processing and transparency of limitations. Evidence regarding model and application testing should be distinguishable from design specifications. The report cites using numbers and refers to publicly available data rather than original farm data collection.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Workflow of the described model includes Python, PyTorch, torchvision, Pillow, NumPy, pandas, and scikit-learn along with other download/data preparation packages. These files are stored on Google Drive. Versions are recorded at runtime in the notebook. No version numbers can be inferred from some website or library releases.

Table 5.1 Technology stack and evidence boundary

Layer	Technology or artefact	Role
Model	PyTorch and torchvision EfficientNet-B0	Transfer learning and seven-category inference
Image preparation	Pillow and torchvision transforms	Orientation, RGB, padding, augmentation and normalisation
Data handling	NumPy, pandas, PyArrow and SciPy	Manifests, parquet source and perceptual-hash processing
Evaluation	scikit-learn and Matplotlib	Classification reports, confusion matrix and curves
Training environment	Colab GPU workflow	Configured training; actual run hardware must come from its log
Persistence	Google Drive and checksum metadata	Prepared-data reuse and resumable epochs
Deployment bundle	inference.pt with model_state and config	Export format in the revised notebook
Application filename	pest_detector.pt	Deployment-facing name stated in the shared record
Mobile and backend frameworks	Not established by the available implementation record	Must be identified from the application repository

5.2 Component implementation

The data preparation program identifies the source categories, generates standardized images, and creates the inventory. Duplication and conflicts are checked before partitioning in groups. The prepped cache contains the manifest file, such that future

sessions will have the same experiment rather than creating a new split each time there is a runtime reset. Checksums help in detecting transfer incompleteness and data mismatch.

Checkpoints contain sufficient information to be able to pick up the experiment from where it left off, including the optimizer and random states. Recovery procedure will verify the dataset identity and training parameters before proceeding. The development report has a CPU training/resume check listed, but this is different from completing the entire Colab schedule [2]. The trained artifact mentioned at PP2 does not have final exported metrics [1] here.

For the inference phase, the backend reassembles the seven-output architecture and imports the exported model_state. The config provides the order of classes and transformations. A general-purpose loading mechanism looking for just the state_dict or model_state_dict will not be compatible with this export. Importation should fail explicitly in case of missing weight or shape mismatches. It must not return predictions from the randomly initialized network.

5.3 Integration with the other components

According to PP2, the prediction workflow is integrated with the NAIMIRIS mobile application [1]. The integration document suggests that a multi-part image analysis endpoint should be used, something like POST /api/pest-detection/analyze. This is an illustrative implementation only and not an independently tested production-ready endpoint. It is recommended that the actual endpoint be extracted from the backend implementation.

5.4 Verification and software testing

Tests are grouped according to data tests, model interface tests, back-end tests, and user interface tests. Table 5.2 details the prioritized test cases. The expected results constitute the acceptance criteria. The execution status of these criteria will be determined when there is documented evidence from the application; this table does not imply that the developed application passes them.

Table 5.2 Core verification cases and expected outcomes

ID	Test and input	Expected outcome
TC01	Supply indices 0 through 6 using a controlled output fixture	Exact names match Table 4.4 without alphabetical reordering.
TC02	Load the exported model_state and config bundle	Seven-output model loads and metadata agrees.
TC03	Remove or corrupt the model file	Service reports unavailable; no random-weight inference.
TC04	Compare notebook and backend processing of one non-square image	Same orientation, 224-square padding and normalised tensor.
TC05	Submit valid RGB, grayscale and transparent images	Supported files convert consistently to RGB without a crash.
TC06	Submit empty, corrupt and oversized files	Clear validation failure before inference.
TC07	Use a synthetic output with top score 0.69	Uncertain status and no chemical guidance.
TC08	Lower score threshold to 0.40 and use top scores 0.49 and 0.40	Margin-only uncertainty with the configurable 0.15 gap rule.
TC09	Use valid scores 0.49 and 0.40 with remaining mass 0.11	Uncertain status; both score and margin criteria fail.
TC10	Use a healthy-category controlled prediction	Healthy-looking wording, monitoring guidance and no disease treatment.
TC11	Use leaf_curl or bbwv2 controlled predictions	Suspected viral condition; no curative-pesticide claim.
TC12	Request guidance for each of the seven keys	Matching versioned content and source identifiers.
TC13	Simulate timeout, failed upload and retry	Clear error and retry; no stale result association.
TC14	Repeat inference with a fixed image on CPU	Stable mapping and finite seven-element output.
TC15	Inspect confidence display and model-metric display	Prediction score is not described as test accuracy.
TC16	Submit unrelated and unsupported-condition images	Record errors and high-confidence failures; do not assume rejection.

Threshold combination warrants consideration. The fact that all scores must add up to one means that a maximum score of 0.70 allows 0.30 maximum for the second place, resulting in the top two difference of at least 0.40. At such defaults, the margin criterion of 0.15 is irrelevant because it does not exclude an accepted prediction. The test is relevant if the thresholds get adjusted. Boundary tests thus confirm the logic used without claiming that the margin offers extra protection in the current context.

Tests with generated outcomes validate the software criteria but not the model's capability to detect illness. The testing of the model is done through the analysis of labeled unobserved images. Guidance tests confirm that lookup and wording are correct, and the agricultural expert review confirms the relevance of the guidance. Each type of evidence addresses a different question and gets a separate log.

5.5 Deployment and operation

Deployment package includes the model chosen, its preprocessing/class data, and the guidance version suitable for that. The startup process checks the 7-output shape, needed configuration and an example inference path. Device used can be either the CPU or CUDA; numerical behavior and latency must be measured on the real machine. One-time loading of the model eliminates costly repetition for each request.

Monitoring in operations needs to detect whether there was an invalid upload, service unavailability, failure during the inference process and prediction uncertainty. The stored data should be enough to troubleshoot any problems without storing images unnecessarily. Changes in deployment demand versioning and a rollback plan. These operational issues remain open for execution verification until the component becomes ready for unrestricted farm use.

6 RESULTS AND VALIDATION

6.1 Research results

The most prominent quantified preparation result in the joint development log is the retention of 4,473 images from a chosen sample of 4,507 images after duplicate/conflict elimination [2]. Also, there is mention of an approximate 102 MB prepared cache and CPU checkpoint and restore check in the same log. These are workflow verification results; not disease classification performance measurements.

Table 6.1 Research evaluation status

Item	Recorded finding or status
Output categories	Six disease-related labels plus healthy; exact mapping documented
Selected inventory	4,507 images before duplicate/conflict processing
Retained preparation total	4,473 reported in the development record; final run manifest not independently audited
Data partitions	Grouped approximate 70/15/15 procedure defined; final image counts not established
Completed training epochs	Not established by a final training log
Held-out accuracy and macro-F1	Not established
Per-class precision recall and F1	Not established
Confusion matrix and error examples	Evaluation export procedure defined; measured outputs not established
Sri Lankan Nai Miris field performance	Not established

6.2 Product and technical results

PP2 Progress Slide of current progress has an EfficientNet-B0 classifier and mobile-connected prediction pipeline showing the uploaded image, class and confidence level [1]. This gives the bounded claim about the prototype implementation that has been reported. There is no recorded response time, software test log or deployment inspection for the above claim.

6.3 User or stakeholder validation

Neither participant testing nor expert validation is documented. Thus, usability and recommendation accuracy are validation challenges to be resolved. Testing for validation purposes should ask participants to upload an image, classify the category and its confidence level, recognize uncertainty and identify the recommended action to take next. Successful task completion and misunderstandings must be distinguished from judgments on aesthetics of the interface design.

Validation by experts should involve evaluation of the interpretation of source labels, disease descriptions and relevance of management recommendations for the specific crop. Special care needs to be taken for leaf curl, anthracnose and BBWV2 because the context may not be sufficient for accurate diagnosis. It must be documented in the test report what expert review was done, under what criteria, any disagreement and subsequent changes without institutional endorsement being provided.

6.4 Integration results

The integration of the mobile to the backend prediction system is stated to be done in PP2, while the integration of the recommendation system is stated to be in progress [1]. A full integration outcome should also show file validation, preprocessing matching, class mapping, guidance lookup, result ownership, and managed failure cases.

6.5 Summary of findings

It provides evidence for the existence of a seven-category EfficientNet-B0 process, a prepared data result and a mobile classification prototype. However, it cannot be considered to provide information about the test accuracy, severity assessment, dosage amount, stakeholders' acceptance or financial benefits at the crop level. Recommendation development is more advanced than the PP2 design, but the implementation of the recommendation is distinct from its design.

7 DISCUSSION

7.1 Interpretation of findings

This project shows the significance of ensuring that the description of the study matches the activity itself. While a classifier with seven outputs is capable of performing image screening of leaves, it cannot prove the claims on localisation and intensity of the earlier detection study. This revised scope ensures that the interface and evaluation standards match.

7.2 Comparison with literature baselines and existing solutions

The published studies involving plant image classifications have indicated that classification could function very well within a particular dataset [3], [4]. The numerical values presented here should be considered to apply only within their contexts. It is not right to compare the performance of NAIMIRIS with the other two methods since their outcomes and the data sets are not the same.

7.3 Research contribution

The contribution of the research is that of an approach to integrating the six labeled disease categories along with a label for healthy appearance within a single classifier pipeline. This involves making choices regarding whether to exclude non-disease labels, how to deal with duplicated classes and the use of validation macro-F1 for the selection process.

7.4 Product contribution

Product contribution is a mobile-based procedure where the input leaf image produces a readable crop health screening result. PP2 captures the core prediction link whereas the subsequent specification reveals how guidance and uncertainty can be incorporated. The stable model keys enable the display wording to be adjusted for Nai Miris without modifying the numerical output order of the model.

7.5 Limitations and threats to validity

The internal validity is compromised due to duplicate images or images that are related, as well as the decision-making process after the observation of test results.

While the grouping procedure is effective at dealing with image-relatedness leakage, it is not able to make plants independent of one another because of the lack of original acquisition identifiers.

The external validity is affected by the mismatch between publicly available chilli/pepper images and Sri Lankan Nai Miris farms. BBWV2, in particular, relies on the use of controlled Capsicum annuum images [14]. There is no local prevalence or performance of a Scotch Bonnet diagnosis from the inclusion of such source.

The problem with construct validity is that it relies on the classification of labels as a sign of a disease. “Leaf curl” is a symptom type with many potential causes. Anthracnose labels need to be validated, and performance in leaf images does not imply performance in identifying diseases in fruits. The criteria for healthy appearance is more restrictive than plant health and confidence is more restrictive than accuracy, being completely irrelevant to severity measurements.

The problem with product validity is that there is no documentation regarding the execution of the test cases as proposed. The recommendation specification is not an analysis of the efficacy of the treatments. It follows that subscription feasibility, decreased pesticide application, and increased yield are proposed benefits.

7.6 Implications

The component can be built upon to support future crop health studies if the evidentiary limits of the component remain explicit. The developers require an accurate export agreement, the researchers require evaluation data, and the farmers require clear language which doesn’t exaggerate the prediction. Meeting the requirements will ensure coherence of the solution.

The following step towards agriculture deployment is the preparation of an expert-reviewed set of local images and evaluation of the current model on this set. The following step towards software deployment is pre-processing and class mapping equivalence. The above steps address specific risks that arise from the current process, rather than introduce extra features into the thesis story.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

Target audience would include Nai Miris farmers and other agricultural investors as well as those giving advice for NAI-MIRIS. The benefit expected to accrue from the proposal is ease in getting initial interpretation and accessibility to related management information. Financial benefits and productivity improvements have not been verified yet.

8.2 Market and competitive position

The target users of Nai Miris are farmers, small agricultural investors, and advisory experts. The suggested positioning is that of a crop-based module integrated with the platform's pre-farming and farming guidance functionalities. The differentiating factor for the positioning is the reliable performance of the module in the local environment, helpful advice, and easy usability as opposed to just an AI tag. There is already a well-established foundation for image classification techniques.

8.3 Business or sustainability model

PP2 recommends using a subscription model for the wider smart farming ecosystem and the possibility of using premium and agriculture partnerships [1]. The pest feature may form part of this wider solution package. There is also a possibility of designing a business model using a free screening and paid advice/recording, but this has not been tested yet.

8.4 Scalability intellectual property and compliance

While backend execution will consolidate model updates, actual scalability is dependent on host capacity, request characteristics, and image transfer cost. Model reuse and limited uploads are valid design decisions, but they do not provide the number of supported users. It is necessary to conduct load analysis and monitoring before wide-scale deployment.

Software copyright and ownership by the development team will not prove any kind of patentable model invention. EfficientNet and other public data sets are covered by their respective licences and authorship requirements. The primary CHILLI source

code is marked with a non-commercial share-alike licence in the research process [2]; hence, licence analysis or separate permission will be necessary before commercial use. Other permissions are required as well.

8.5 Contribution to the Sustainable Development Goals

There is possible relevance of the project to food-production resilience, technology accessibility, and responsible use of resources. The above three points relate to SDGs 2, 9, and 12 [18] respectively. It is an intended connection rather than an actual achievement of the SDGs. An advisory system that provides unreliable information could negate the intended outcome.

Table 8.1 Intended sustainability contribution and evidence needed

Theme	Potential contribution	Evidence needed
SDG 2	Earlier attention to crop-health problems	Local accuracy and timely-action study
SDG 9	Accessible digital agricultural support	Usability, availability and deployment evidence
SDG 12	More deliberate management decisions	Review of advice and observed input-use outcomes
Resource efficiency	Reuse of prepared data and checkpoints	Measured training and storage costs

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis discusses the AI-Based Pest Detection and Control System, which is an image classifier with seven categories as part of the NAIMIRIS framework. The chosen architecture is EfficientNet-B0 using PyTorch and is fine-tuned through transfer learning for six classes of diseases and one class for healthy crops.

9.2 Achievement of objectives

Table 9.1 distinguishes documented work from outstanding validation. Achievement is assessed against the revised scope, while deferred original objectives remain explicitly identified.

Table 9.1 Objective achievement and evidence status

Objective	Status	Basis and remaining work
Assemble seven-category public data	Partially evidenced	Sources and reported preparation totals exist; final run manifest requires reconciliation.
Implement reproducible preparation	Documented workflow	Duplicate controls, groups and cache specified; final artefacts should accompany evaluation.
Adapt EfficientNet-B0	Configuration and PP2 implementation reported	Final checkpoint history and held-out report remain outstanding.
Integrate mobile prediction	Reported at PP2	Complete integration execution evidence remains outstanding.
Provide reviewed management guidance	Design specified	Integration and local expert validation remain outstanding.
Evaluate accuracy usability and performance	Incomplete	Test cases are defined; final empirical outcomes are not established.
Original severity and dosage objectives	Deferred	Not produced by the seven-category classifier.

9.3 Individual research and product contributions

The individual research contribution is the classification experiment that can be traced and the requirements for evidence. Individual Product Contribution

The individual product contribution is the implementation and incorporation of the crop health prediction component in NAIMIRIS. The use of stable classes mapping, explicit preprocessing contract and separate guidance layer makes this component more analysable.

9.4 Future directions

The first task is to restore the experiment manifest, checkpoints, histories, evaluations, and report per-class results and errors. Before drawing more conclusions on accuracy, the backend must be validated against the padding and model_state interface described in the notebook. After that, the main test cases should be performed with evidence versioning.

A collection of Sri Lankan Nai Miris images curated by experts will be necessary for the assessment of transfer of crops and environments. Whenever possible, different independent plants, farms and seasons must be included. Source balanced and leave source out approaches may help assess the background dependence. Comparisons against simpler models can be helpful in selecting the model.

More research may involve calibration of prediction score, assessment of unsupported image handling, and validation of the algorithm with experts in agriculture and local farmers. For severity estimation, a new task with labels will have to be created and evaluated separately. Extension of edge inference, multilingual support, and long-term observation are potential future directions.

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GLOSSARY

Term	Meaning in this thesis
Nai Miris	The intended crop context of the NAIMIRIS Scotch Bonnet application; not a relabelling of all public source crops.
Image classification	Assignment of one category to a whole submitted image.
Confidence score	A model output score for a prediction; not measured accuracy or disease severity.
Transfer learning	Adaptation of pretrained features to the seven-category task.
Grouped split	Partitioning related-image groups together to reduce leakage.
Data leakage	Information shared between development and evaluation data that makes performance misleading.
Domain shift	Differences between source images and the intended deployment environment.
Knowledge base	Versioned category-specific management content separate from the neural network.
Severity estimation	Measurement or classification of affected extent; outside the current classifier.
Closed-set classifier	A model that assigns inputs among a fixed list even when the real condition is unsupported.

APPENDICES

Appendix A Research instruments and reproducibility records

The additional documents substantiate the methodologies and test procedures detailed in Chapters 3 and 5. These provide the information needed to replicate the experiment and conduct the application tests. They are not meant to replace the principal documentation or the finished empirical findings. The division of results of the research, product testing, and supporting appendices as per the university guidelines remains [19].

Table A.1 Reproducibility artefact register

Artefact	Evidence carried
Source configuration	Dataset identifiers, versions, licences and selection rules
manifest.csv	Image path, source, label, class ID, group and partition
Duplicate and conflict audits	Removed exact duplicates and quarantined label conflicts
Prepared cache metadata	Configuration identity and archive/manifest checksums
Epoch checkpoint and commit record	Model and optimizer states, epoch, checksum and dataset identity
Training history and versions	Actual completed epochs, metrics and software environment
classification_report.csv	Per-class precision, recall, F1 and support
test_predictions.csv	True class, predicted class and seven scores for each test image
metrics.json	Final test metrics and selected epoch
inference_config.json and inference.pt	Deployment mapping, preprocessing and model_state
Application test log	Build/model versions, inputs, expected and actual outcomes

Table A.2 Detailed acceptance fixtures for selected test cases

Case	Procedure and fixture	Required evidence and status
TC01	Inject each index in turn and compare model key and display label with Table 4.4.	Seven mapping assertions; execution pending.
TC02	Load the notebook export using model_state and config; check output width seven.	Startup log and sample output shape; execution pending.
TC04	Use a portrait image with EXIF rotation and visible edges in notebook/backend.	Pixel/tensor comparison and tolerance; execution pending.
TC07	Use valid scores [0.69, 0.10, 0.07, 0.05, 0.04, 0.03, 0.02].	Uncertain response with no chemical guidance; execution pending.
TC09	Use valid scores [0.49, 0.40, 0.03, 0.03, 0.02, 0.02, 0.01].	Score/margin checks recorded; execution pending.
Threshold boundary	Test a valid top score of exactly 0.70, then values immediately below and above.	Equality behaviour matches documented less-than rule; execution pending.
Configurable margin	Set score threshold to 0.40 and margin to 0.15; use the TC09 distribution.	Margin-only deferral under changed settings; execution pending.
TC10 and TC11	Inject healthy, leaf_curl and bbwv2 results into guidance retrieval.	Correct warnings and absence of curative claims; execution pending.
TC16	Use unrelated objects, non-target leaves and unsupported symptoms.	Full outcome log including high-score failures; execution pending.
Recovery	Interrupt after a committed epoch and corrupt a separate test copy of a later checkpoint.	Resume/fallback log without changing the original run; execution pending.

Appendix B Academic integrity and AI use disclosure

The generative AI assistance has been employed for help with structuring, language generation, identifying inconsistencies, interpreting the shared development discussion and preparing test case specifications. The shared discussion also includes AI-assisted data set selection, notebook integration and backend integration. The updated thesis differentiates between those test case specifications and experimental results measured independently.

The student is the author of the final research report, owns the dataset rights, is responsible for the implementation and has all the necessary citations. Test results, participant results and application pass/fail results need the proper research report. The disclosure statement does not state that the test cases not run have passed or that the guidance generated by AI has been approved by an agriculture expert.

Appendix C Disease descriptions and management knowledge base

Below is listed the content to be included in the interface of Nai Miris. The entries maintain the original source-label relationship in the model and must be examined by an agricultural expert before deployment. The crop-related display language is a modification of the application and is not a guarantee that all of the training images or diagnoses belong to Scotch Bonnet.

C.1 Healthy looking Nai Miris leaf

The healthy classification means that there is similarity to images of healthy crops. In the case of this classification, the user should be advised on further observation as well as proper crop management, but there should be no statement about the plant being free of diseases throughout its body. The user with persisting symptoms can upload another relevant image.

C.2 Bacterial spot of Nai Miris

The bacterial spot disease can attack both pepper leaves and fruit, and the symptoms might mimic other diseases. It is important that the recommendation emphasize the use of healthy seeds, cleanliness of the tools, no working when the foliage is wet, and avoidance of splashes between plants. According to the guidelines by the University of Minnesota, copper cannot give full protection under all conditions [8].

C.3 Cercospora leaf spot of Nai Miris

This is an example of the category Source as it refers to Cercospora leaf spot rather than all types of leaf spot. Pale centers and dark edges could provide the basis for suspicion, but alone they don't prove causality. Information on management should focus on the elimination of diseased material and proper spacing as well as leaf spot management information [10].

C.4 Suspected leaf curl disease of Nai Miris

The training key leaf_curl is based on the source group whose name is Curl Virus. The user interface must respect the provenance of the key while at the same time informing him that curling, distortion, and stunting may result from various causes. It must instruct the user to look at the plant for proof and not to assume the curling means a

certain viral infection. Management of the vector does not cure the plant but controls its spread.

C.5 Powdery mildew of Nai Miris

Pepper powdery mildew causes light-colored upper surface patches and a powdery development on the underside of leaves. Early examination is recommended, and basic sanitation alone does not provide sufficient protection from the disease [9]. The information requested should include the correct photos and direct the person to relevant management practices. Information on foreign materials is not translated for automatic instructions in Sri Lanka.

C.6 Suspected anthracnose of Nai Miris

Anthracnose infects parts of the pepper plant and is caused by fungi of the genus Colletotrichum. In terms of DOA instructions in the locality, symptoms of anthracnose on the leaf, stem, and pod, as well as the importance of using clean seeds, sanitation, drainage, and elimination of infection are highlighted [7]. The interface could prompt a specific examination of the fruit, although the current image classifier for the leaves is not verified for fruit lesions.

C.7 Suspected BBWV2 infection in Nai Miris

BBWV2 is still broad bean wilt virus 2, despite being illustrated in Nai Miris. The data from which this virus has been identified comes from laboratory-infected C. annuum [14]. This does not prove that this virus is prevalent locally or its reliable detection in Scotch Bonnet. Fabaviruses are transmitted by aphids in a non-persistent manner [17]. The guidance should focus on getting experts to confirm the virus, as well as minimizing infection sources and vectors.

C.8 Shared guidance rules

A source identifier, revision date, and review status should accompany each listing. In case of doubtful results, no chemical advice is given, and severe or progressive symptoms require medical consultations. Brand name, rate, interval, and pre-harvest intervals are never fabricated by the model. Future chemical advice will always have to be accompanied by current registration and label checks in the local area.